

Philosophical Logic Task D

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Consider the following sentence: $(\sigma) \forall x(@Fx \rightarrow Fx)$.

(a) Specify an idiomatic English sentence σ_{ENG} that translates σ using the following dictionary:

F : ...is valuable

Response: All the valuable objects are valuable.

(b) Show that:

i. $\models_{2D} \sigma$

Proof. Assume for reductio that $V_{\mathcal{M},g}^2(\forall x(@Fx \rightarrow Fx), w, w) = 0$. Then $V_{\mathcal{M},g_{d'}}^2(@Fx \rightarrow Fx, w, w) = 0$ for some $d' \in \mathcal{D}$. Hence, $V_{\mathcal{M},g_{d'}}^2(@Fx, w, w) = 1$ and $V_{\mathcal{M},g_{d'}}^2(Fx, w, w) = 0$. But $V_{\mathcal{M},g_{d'}}^2(@Fx, w, w) = 1$ implies $V_{\mathcal{M},g_{d'}}^2(Fx, w, w) = 1$. So, both $\langle [x]_{g_{d'}}^x \rangle \notin \mathcal{J}(F)$ and $\langle [x]_{g_{d'}}^x \rangle \in \mathcal{J}(F)$, a contradiction ■

ii. $\not\models_{2D} \Box\sigma$

Proof. Our countermodel is

$$\begin{aligned}\mathcal{W} &= \{w_1, w_2\} \\ \mathcal{D} &= \{d\} \\ \mathcal{J}(F) &= \{\langle d, w_1 \rangle\}\end{aligned}$$

■

iii. $\models_{2D} F@ \sigma$

Proof. AFC that $V_{\mathcal{M},g}^2(F@ \forall x(@Fx \rightarrow Fx), w, w) = 0$, which implies $V_{\mathcal{M},g}^2(@ \forall x(@Fx \rightarrow Fx), v, w) = 0$ for some $v \in \mathcal{W}$. So, $V_{\mathcal{M},g}^2(\forall x(@Fx \rightarrow Fx), v, v) = 0$, which implies $V_{\mathcal{M},g_d}^2(@Fx \rightarrow Fx, v, v) = 0$ for some $d \in \mathcal{D}$.

Hence, we have both $V_{\mathcal{M},g_d}^2(@Fx, v, v) = 1$ which implies $V_{\mathcal{M},g_d}^2(Fx, v, v) = 1$ and $V_{\mathcal{M},g_d}^2(Fx, v, v) = 0$. So, both $\langle [x]_{g_d}^x \rangle \notin \mathcal{J}(F)$ and $\langle [x]_{g_d}^x \rangle \in \mathcal{J}(F)$, a contradiction ■

(c) Assess what support, if any, these results give to the following claims (1000 words, total)

i. What σ_{ENG} is knowable a priori.

Response: To give a rough characterisation as what is meant by a priori, we might say that a sentence σ is knowable a priori if it can come to know of σ via reflection by the armchair, or without any empirical evidence. We can identify a statement as knowable a priori iff $\models \sigma$ i.e. it is a logical tautology. In this case, the fact that σ is a 2D-QML tautology might allow us to say that, we can derive σ_{ENG} without the use of any empirical evidence and by pure proof. But the objection by Kripke employs the result of b(ii) i.e. that $\not\models_{2D} \Box\sigma$ which implies that it is not knowable a priori that it is necessarily the

case that σ . So, this means that there are worlds, call them w in which σ is false. So, the objection runs, how do we know that our world is not such a world where σ is false except via empirical evidence?

But the reconciliation, according to Evans and Davies is the following. First, inhabitants of w , when uttering σ_{ENG} , actually refers to the actual world $w_{@}$ and its objects i.e. *the* valuable objects. (While they may not know what their english sentence refers to i.e. the actual valuable objects currently in the actual world, and so may not come to know σ a priori, σ_{ENG} nevertheless remains knowable a priori by suitable semantic argument). And since we know that by b(i) that $\models_{2D} \sigma$ is valid at every pair of reference-evaluation worlds $\langle w, w \rangle$, we can infer that this holds at the actual world $\langle w_{@}, w_{@} \rangle$. So, while by b(ii), σ_{ENG} is contingent i.e. falsity at a pair of distinct worlds $\langle w_{@}, w \rangle$, we can still know that σ_{ENG} is true in our world a priori, from the standpoint of our actual world. So, Kripke's puzzle is a non-issue – contingency and a priori do not come into conflict.

- ii. What σ_{ENG} says is only contingently the case.

Response: What makes something contingently the case? We might say that a sentence s is contingent iff it is possible for s to be false in some possible world. The intuition is clear: if contingent means that it is possible that s is either true or false, then we can define this as just being true in one possible world and false in another. To this end, b(ii) provides support for the fact that σ_{ENG} is a contingent fact since $\not\models_{2D} \Box \sigma$.

However, Evans and Davies point out that there are two kinds of contingencies. The first is superficial contingency i.e. that truth is merely with respect to worlds and the contingency is determined wholly by $\Box$ s or modal operators. There is nothing about the worlds which σ_{ENG} is evaluated against that renders the statement true or false. In contrast, deep contingency is determined by the modal properties of sentences, rather than the modal operators themselves. This is because a deeply contingent sentence would be made true or false by the state of affairs *in* the world in which it is evaluated against. So, what $\not\models_{2D} \Box \sigma$ is merely that σ_{ENG} is superficially contingent – its contingency is merely by virtue of the $\Box$ operator. However, $\models_{2D} F@ \sigma$ suggests that σ_{ENG} is also deeply necessary because whichever possible world were to be considered the actual, σ_{ENG} would still be true and hence does not depend on any contingent feature of reality. More clearly, I think the intuition is this: σ_{ENG} is only superficially contingent because if I utter σ_{ENG} in world $w_{@}$, and σ_{ENG} was taken out of context to be about $w \neq w_{@}$, then it is very clearly the case that σ_{ENG} is contingent on what worlds I evaluate σ_{ENG} against. But this does not account for the fact that whether my utterance of σ_{ENG} is true depends on the world in which I am situated in. Something that is deeply necessary or contingent realises that the world in which I evaluate a sentence against must always be the same world in which I utter it from (which may move and is not fixed). So, σ_{ENG} is not deeply but only superficially contingent.

- iii. The truth of an utterance of σ_{ENG} does not make any substantive demand of the world.

Response: Let us say that the truth of an utterance of some sentence s makes a substantive demand of the world if it is the case that some state of affairs in some world w must be true for s to be true in w . We have established earlier that σ_{ENG} is a deeply necessary sentence since $\models_{2D} F@ \sigma$, and so it is true along the diagonal which basically means that in whichever world σ_{ENG} is uttered, σ_{ENG} will come out true. The corollary of this is that the truth of my utterance of σ_{ENG} does not depend on any contingent feature of reality, and so *prima facie*, does not require any state of affairs to obtain for σ_{ENG} to be true. In other words σ_{ENG} is true no matter what.

But what gives me pause in arguing that this conclusion follows decisively are two things. First, notice that the definition of *deeply* necessary, as put forward by Davies, is that it depends on truth *in* a world. If deep necessity depends on the world beyond mere internal theoretical features of semantics, then does it not make some substantive demand of the world? I think this can be reconciled by clarifying what Davies actually means by a sentence being made true in a world. I suspect he what he meant to say is that a sentence σ can be made true in a world in virtue of it *being uttered* in the world, rather than some state of affairs obtaining. So, the truth of the utterance of σ_{ENG} does not make a substantive demand on the world. The second point is more general, if not orthogonal to the discussion – but Davies points out that a priority might not entail deep necessity. In other words, $\models_{2D} \phi$ might not entail $\models_{2D} F@ \phi$. This is because any reductio proof against the opposite claim that a deeply contingent

but a priori claim is impossible would return to the claim to show that a modally contingent claim S implies that we cannot be a priori epistemically guaranteed that S does obtain. All in all, this second point is orthogonal to the discussion of σ_{ENG} making a substantive demand on the world, but it serves to warn us that we cannot assume that its truth of utterance does not in virtue of it being a priori.